

Element E: Application of STEM

E1 - Application of STEM in your community project.

Our proposed solution includes the creation of an artificial window that is supported by areas of STEM. It mostly focuses on engineering and includes some aspects of science (psychology). With our project, we will use our engineering knowledge to assemble the frames and mechanical components of the work. Our goal is to be able to make our project stand out and have a psychological effect on people, by using psychological theories to understand our customer's wants and needs. Through the manipulation of art, we will attempt to make an impact with our project.

E2 - How STEM is used to support the design solution.

- While E1 was focused on how STEM applies to the design requirements, E2 is focused on how STEM is used to support the design solution.
- Use STEM concepts to justify your design decisions. It's important that the STEM Principles and Practices that are discussed are actually appropriate and useful to the problem.

E3/4 - Expert/ Mentor Review

Our field expert is An's dad, Mr. Le. Mr. Le's experience in building and construction is evident in his personal and work life. Our team member mentioned that her room was mostly constructed and designed by her dad, which shows his prevalent background in assembling and decorating, our main focus in this project. He also has experience in wood work and electrical engineering.

Interview Questions and Transcript:

Question: What STEM related fields were you involved in?

Answer: "Ever since I was a kid, I've always had an interest in building things, especially anything woodworking. I learned how to do plenty of things from my dad, such as woodworking, electrical things, and even concrete. I also had a job being a silversmith as a child, this is where I learned skills such as welding. If I had to identify with a category in stem it would definitely be engineering."

Question: Which STEM category do you believe best identifies with our project? Do you have experience in this category?

Answer: "I believe that this project mostly identifies with engineering, since building the window will involve a lot of wood working. I've done plenty of at home projects that involved wood cutting. I've made shelves, stands for fish tanks, and even a gazebo."

Question: Our project involves woodworking, do you have any advice?

Answer: "Measure twice, cut once. Be careful with your tools, don't throw them around. Your workspace should be organized and keep a close eye on your sharp tools. Wear gloves because

there are a lot of sharp edges and it gives you a grip on the wood. Mark measurements with a sharp pencil. And most importantly, don't rush."

Question: How heavy are you expecting the project to be? Do you think the magnets will hold this weight?

Answer: "I predict that the project will weigh around 5-7 lbs, including the light box, wood frame, and other components. The magnets are strong, so they should be able to withstand the weight if it's not too heavy. If the project is about 10 lbs, the magnets might not be strong enough to hold that much."

Question: How do you think we should assemble the window? Nails, glue, or pegs?

Answer: "I would recommend glue and nails. The glue can be used for smaller aspects of the project for example the portrait and magnets. But for assembling the wood parts together I recommend nails because it'll give the project a more sleek finish. With screws, the head will pop out and make the window unappealing."

Question: What's the most important thing that you should keep in mind when designing, in your opinion?

Answer: "When you design something, don't think your initial idea is the final. While working, you will find improvements and you should act on those. Don't be stubborn and be open minded to change."

E5- Application of STEM

Mr. Le told us that the magnets' strength is enough to hold up to 5-7 lbs, which informed our decision to reconsider the additional features such as the planter box. While the idea of a planter box may be appealing to our customer, it would not be ideal considering the heaviness of the project. In addition, Mr. Le suggested using glue and nails to assemble the pieces of our project together. This verified our decision to use hammers, screwdrivers, and other hand-held tools to ensure the durability of our project.
